

MECH 191 — Metallurgy

Semester Alloy Research Portfolio — Instructor Answer Key

Fall 2026 • Leeward Community College • Instructor: Kai Santos

How to Use This Key

This document mirrors MECH191_Modules_ResearchBased.docx — Modules 1–15 in order, Step 1 (Research & Documentation), Step 2 (Short-Answer), and Step 3 (Critical Thinking). It is written to guide grading, not to be handed to students.

Because this is a research-based portfolio, most short-answer and critical-thinking questions do not have one single "correct" numeric answer — students may cite different real, published sources and reach slightly different values (a MatWeb entry and an AZoM article for the same alloy can differ by a few percent). Award credit for any answer that cites a real, plausible published value for the correct alloy/condition and reasons soundly from it, even where the model answer below uses a different representative figure. Treat a value that is wildly outside normal published ranges for that alloy as a flag to check the student's source, not automatically as wrong.

The exception is the Hardness Log questions (Modules 6, 9, 10, 12, 13, 14): Appendix A gives a single fixed reference HV for each specimen, so the d_{avg} calculated from $HV = 1.854 \times F / d^2$ does have one correct numeric answer, shown below. A student who cites a different published HV for the same alloy and shows correct algebra from that value should still receive full credit — note the discrepancy rather than marking it wrong.

For Step 1 (Research & Documentation), there is no separate answer key entry — it's an open research task graded on completeness and correct citation. A short "Representative Data" note is included per module below as a grading reference, showing the kind of published values a well-researched answer should land on.

The portfolio's own Grading Summary (reproduced below) governs point allocation: 5 pts/module × 15 modules = 75 pts, plus a 25-pt holistic end-of-semester review = 100 pts total.

Grading Summary

Research documentation — 1 pt — Credit for each data point researched and properly recorded, with a real source cited

Short-answer questions (Step 2) — 2 pts — Credit apportioned across all questions in the set

Critical-thinking questions (Step 3) — 1 pt — Equation applied correctly and reasoning tied to real published data

Source citation — 1 pt — Publication/database name and specific value or table referenced, recorded for each data point used

Total per module — 5 pts — 15 modules × 5 pts = 75 pts

End-of-semester portfolio review — 25 pts — Holistic; all 15 modules as a complete document

TOTAL — 100 pts

MODULE 1 · WEEK 1 | Atomic Structure & Bonding (Specimen A — 1018 steel, normalized)

Representative Data (Step 1): 1018 normalized: ~0.18% C, ~0.6–0.9% Mn; yield strength $\approx$ 220–260 MPa, tensile strength $\approx$ 400–440 MPa, elongation $\approx$ 25–30%; typical published ASTM E112 grain size number for normalized 1018 falls around G7–G8.

Step 2

Q1. Normalized 1018 is expected to show a roughly **equiaxed** grain structure — grains approximately equal in all directions. Normalizing (austenitizing above the upper critical temperature, then air-cooling) allows full recrystallization into new, uniform, strain-free grains, erasing any elongated/banded structure left over from prior hot rolling.

Q2. Metallic bonding is **non-directional**: positive ion cores are held together by a delocalized "sea" of shared electrons rather than fixed, directional bonds like covalent or ionic solids have. Because the bonding isn't tied to a specific bond angle or neighbor, atoms at a grain boundary — where the lattice is disordered and misoriented relative to each neighboring grain — can still be held together by that shared electron sea. The boundary is higher-energy than the bulk lattice but not a broken bond, so the material doesn't crack simply because a boundary exists.

Q3. Using a typical published ASTM E112 grain size number of **G $\approx$ 8** for normalized 1018: the ASTM E112 chart gives a nominal average grain diameter of about **22 μ m (0.022 mm)** for G8 (G7 would give roughly 32 μ m). Accept any answer using a published G-number in the 6–9 range with the diameter that number implies.

Q4. **MnS inclusions** in rolled/normalized 1018 are typically elongated along the rolling direction. They create mechanical anisotropy (lower ductility and impact toughness transverse to the rolling direction than longitudinal), but their softness and controlled distribution also improve machinability by helping chips break cleanly.

Step 3 — Critical Thinking

C1. Sample X (finer grain) is expected to have the **higher yield strength**. Hall-Petch ($\sigma_y = \sigma_0 + k_y/\sqrt{d}$) shows yield strength rising as grain diameter d falls. Physically, more grain-boundary area per unit volume means more obstacles that block and pile up dislocations; a higher applied stress is needed to push dislocations across more, closer-spaced boundaries and activate slip in the next grain.

C2. No — strength cannot rise without limit as $d \rightarrow 0$. In real metals, once grain size drops into the nanocrystalline range (roughly below 10–15 nm), there isn't enough room inside a grain to pile up dislocations at all, and deformation shifts to grain-boundary sliding and diffusion instead of dislocation motion — strength actually levels off or falls again (the "inverse Hall-Petch" effect). Practical processing limits on how fine a grain size can be produced are a real-world limit as well.

MODULE 2 · WEEK 2 | Mechanical Properties (Specimens A — 1018, B — 1045, both normalized)

Representative Data (Step 1): 1045 normalized: ~0.45% C, ~0.6–0.9% Mn; published hardness $\approx 160\text{--}180$ HV (Appendix A reference: 170 HV); tensile strength commonly cited around 620–670 MPa for normalized 1045, though some published sources report values as high as ~750 MPa depending on section size and exact cooling rate.

Step 2

Q1. Specimen B (1045) has **more pearlite** than A (1018). 1045's ~0.45% C is roughly 2.5× 1018's ~0.18% C; by the lever rule, higher carbon content increases the equilibrium pearlite fraction in a normalized hypoeutectoid steel.

Q2. Using the Appendix A reference HV for B (170) and $HV = 1.854 \times F / d^2$, with $F = 1$ kgf: $d_{\text{avg}} = \sqrt{(1.854/170)} = \sqrt{0.01090} \approx \mathbf{0.104\text{ mm}} (\approx \mathbf{104\text{ }\mu\text{m}})$.

Q3. Refining grain size by 50% ($d \rightarrow 0.5d$) increases the $k_y/\sqrt{d}$ term by a factor of $1/\sqrt{0.5} \approx 1.41$ — so yield strength **increases** (qualitatively, by a meaningful margin, though the exact increase also depends on σ_0 and k_y for that steel).

Q4. Specimen B (1045) is expected to have **higher tensile strength**, consistent with its higher pearlite fraction — pearlite (alternating ferrite/cementite lamellae) is harder and stronger than the softer, more ductile proeutectoid ferrite that dominates in low-carbon 1018.

Step 3 — Critical Thinking

C1. The **ferritic** specimen is expected to show more strain (elongation) at the same engineering stress. Ferrite is the softer, more ductile phase — it yields at a lower stress and can accumulate substantially more plastic strain before reaching a given stress level than the harder, less ductile pearlitic lamellar structure, which resists dislocation motion more strongly and reaches the same stress at lower strain.

C2. Past the yield point, dislocations begin gliding extensively through the crystal lattice, multiplying and interacting/tangling with one another (strain hardening). This permanent, irreversible rearrangement of atomic planes is fundamentally different from the reversible elastic stretching of atomic bonds that Hooke's law describes, so stress and strain are no longer simply proportional.

MODULE 3 · WEEK 4 | Carbon & Alloy Steels (Specimens A — 1018, B — 1045, C — 1075, all normalized)

Representative Data (Step 1): 1075 normalized: ~0.75% C, ~0.4–0.7% Mn; published hardness for normalized 1075 commonly falls around 200–240 HV (Appendix A reference: 220 HV). The eutectoid composition on the iron-carbon diagram is ~0.76 wt% C at ~727°C.

Step 2

Q1. On the iron-carbon diagram, all three grades fall in the hypoeutectoid region (left of 0.76% C): 1018 (~0.18% C) sits well to the left, showing mostly proeutectoid ferrite with scattered pearlite islands; 1045 (~0.45% C) sits roughly in the middle; 1075 (~0.75% C) sits almost exactly at the eutectoid line, so its room-temperature structure is nearly all pearlite with only a thin ferrite network.

Q2. Using Fraction pearlite $\approx (C_0 - 0.022) / (0.76 - 0.022)$: 1018 $\rightarrow (0.18 - 0.022) / 0.738 \approx \mathbf{0.21 (21\%)}$; 1045 $\rightarrow (0.45 - 0.022) / 0.738 \approx \mathbf{0.58 (58\%)}$; 1075 $\rightarrow (0.75 - 0.022) / 0.738 \approx \mathbf{0.99 (99\%)}$. Ranked: **1075 > 1045 > 1018**.

Q3. Pearlite lamellar spacing is set by the undercooling/cooling rate during transformation — a faster cool produces finer, more closely-spaced lamellae. Since all three grades are normalized (same air-cooling process), they transform at similar rates and develop similar lamellar spacing. Carbon content instead controls *how much* pearlite forms (via the lever rule), not the fineness of the lamellae within it.

Q4. No — hardness does not scale 4×. The Appendix A reference values (A = 125, B = 170, C = 220 HV) show roughly a 1.76× increase, not 4×. Pearlite fraction itself doesn't scale linearly with carbon content either (21% $\rightarrow$ 58% $\rightarrow$ 99%, a ~4.6× increase) and, more importantly, ferrite already contributes a baseline hardness on its own — hardness is closer to a weighted blend of the two phases' hardness values than a value driven purely by carbon content.

Step 3 — Critical Thinking

C1. Yes — increasing carbon from 1018 to 1045 increases pearlite fraction from ~21% to ~58% per the lever-rule calculation above, which matches published microstructure descriptions: 1045 is consistently described as showing substantially more pearlite than 1018.

C2. Above the eutectoid composition (hypereutectoid steel), the equation no longer applies because the microstructure changes character: instead of more pearlite forming, **proeutectoid cementite** forms first at prior-austenite grain boundaries during cooling (rather than proeutectoid ferrite), and the remaining austenite still transforms to pearlite at exactly the eutectoid composition — so the structure becomes pearlite plus a grain-boundary cementite network, never more than 100% pearlite.

MODULE 4 · WEEK 5 | Stainless Steel Families (Specimens D — 304, E — 316, F — 17-4PH)

Representative Data (Step 1): 304: ~18% Cr, ~8% Ni, austenitic (FCC). 316: ~16–18% Cr, ~10–14% Ni, ~2–3% Mo, austenitic (FCC). 17-4PH: ~15–17.5% Cr, ~3–5% Ni, ~3–5% Cu, ~0.15–0.45% Nb, martensitic matrix strengthened by aging (Cu-rich precipitates), typically supplied in an H900–H1150 aged condition.

Step 2

Q1. FCC metals have close-packed {111} slip planes with relatively low stacking-fault energy, giving a low-energy pathway for a stacking fault (twin) to form during annealing rather than requiring full dislocation dissociation. Martensitic 17-4PH's body-centered tetragonal/BCC-derived structure doesn't offer that same low-energy twinning pathway under normal annealing/aging conditions, so annealing twins are essentially absent.

Q2. Molybdenum reinforces the passive chromium-oxide film, particularly against chloride attack — it forms stable molybdate species within the passive layer that resist local breakdown, substantially raising resistance to pitting and crevice corrosion in chloride-rich environments like seawater.

Q3. Specimen F's microstructure — a martensitic matrix with fine, aging-induced Cu-rich precipitates — is fundamentally different from D and E's austenitic, twinned equiaxed grain structure. This shows that "stainless" describes a corrosion-resistance category (achieved by $\geq 10.5\%$ Cr forming a passive Cr_2O_3 film), not a single microstructure — depending on composition and heat treatment, a stainless steel can be ferritic, austenitic, martensitic, duplex, or precipitation-hardening, unlike a single carbon-steel grade whose structure stays within the ferrite-pearlite family regardless of carbon content.

Q4. 17-4PH is expected to show **less** work hardening from the FCC-twinning mechanism specifically, because that mechanism doesn't operate in its martensitic structure. Austenitic 304, by contrast, notoriously work-hardens heavily during machining via strain-induced twinning/transformation, which is part of why it's considered a difficult-to-machine grade. 17-4PH's own strength instead comes from a matrix that's already hardened by aging, giving it a more predictable, lower work-hardening response during cutting.

Step 3 — Critical Thinking

C1. Using $\text{PREN} = \% \text{Cr} + 3.3(\% \text{Mo}) + 16(\% \text{N})$: a typical ~2.5% Mo addition in 316 is worth roughly $2.5 \times 3.3 \approx 8$ **PREN points**. The formula weights Mo so heavily because molybdenum's contribution to stabilizing the passive film against localized/chloride attack is disproportionately large per weight percent compared to chromium's baseline passivation effect.

C2. PREN remains a useful qualitative comparison across all three (it tracks pitting resistance from Cr/Mo/N content in any stainless), but it's an incomplete comparison for 17-4PH specifically — that grade's overall corrosion performance also depends heavily on its aging condition, precipitate distribution, and martensitic matrix (including susceptibility to hydrogen embrittlement and sensitization at certain aging temperatures), none of which PREN captures. PREN is best treated as a general pitting-resistance indicator, not a complete corrosion-performance comparison, for a precipitation-hardening martensitic grade.

MODULE 5 · WEEK 6 | Nonferrous Metals & Phase Diagrams (Specimens G — 6061-T6, K — 70/30 brass)

Representative Data (Step 1): 6061: ~0.8–1.2% Mg, ~0.4–0.8% Si (forms Mg_2Si); T6 = solution heat treated + artificially aged. 70/30 brass (260): ~70% Cu, ~30% Zn, single-phase alpha (FCC).

Step 2

Q1. Specimen K (70/30 brass) is classically known for very prominent, wide annealing twins — brass has an unusually low stacking-fault energy (roughly 5–10 mJ/m^2 for 70/30 brass), even lower than austenitic 304 stainless (roughly 20–40 mJ/m^2), so K's twins are typically more numerous and visually wider than D's.

Q2. Aging precipitates fine Mg_2Si particles out of a supersaturated solid solution. These fine precipitates impede dislocation motion (Orowan/precipitation strengthening), increasing hardness and strength well above the as-quenched (solution-treated only) condition.

Q3. 6061-T6's Mg_2Si precipitates are very fine — nanometer-scale (GP zones and the metastable β'' phase), too small to resolve optically — and densely, uniformly distributed throughout the aluminum matrix. Dislocations must bow around or between them (Orowan looping), which is what raises the alloy's strength.

Q4. Solution treatment (heating into the single-phase α field, roughly 530°C) dissolves Mg_2Si into a homogeneous supersaturated solid solution; quenching traps that supersaturation at room temperature. Aging then allows controlled, fine-scale precipitation of Mg_2Si . Skipping solution treatment leaves coarse, incoherent Mg_2Si particles already present from casting — aging alone cannot redissolve and re-precipitate them into the fine, coherent form needed for strengthening, so the alloy stays soft (similar to an O-temper).

Step 3 — Critical Thinking

C1. The lever rule requires the alloy to sit inside a **two-phase field** at the temperature of interest, with a horizontal tie-line connecting two known equilibrium-phase compositions at that temperature — the rule reads off the relative amounts of those two specific phases from where the overall composition falls along that tie-line.

C2. The lever rule only tells you the equilibrium **weight fractions** of two coexisting phases at a given temperature and composition — it says nothing about precipitate size, spacing, coherency, or distribution, which is what actually determines the strengthening mechanism and mechanical properties in a precipitation-hardened alloy like 6061-T6, or the solid-solution strengthening in brass. Two alloys with identical lever-rule phase fractions could have very different strengths depending on those microstructural details.

MODULE 6 · WEEK 7 | Mechanical Testing (Specimens A — 1018, B — 1045)

Representative Data (Step 1): Confirm HV values against a second source where possible; published typical grain size for normalized 1018/1045 commonly cited in the G7–G8 range.

Step 2

Q1. Using $HV = 1.854 \times F / d_{avg}^2$ ($F = 1 \text{ kgf}$) and Appendix A's reference values: A (125 HV) $\rightarrow d_{avg} = \sqrt{(1.854/125)} = \sqrt{0.01483} \approx \mathbf{0.122 \text{ mm (122 } \mu\text{m)}}$. B (170 HV) $\rightarrow d_{avg} = \sqrt{(1.854/170)} \approx \mathbf{0.104 \text{ mm (104 } \mu\text{m)}}$. Both become the first two entries in the Hardness Log.

Q2. Using a typical published grain size number of $G \approx 8$ for normalized 1018: $N = 2^{(8-1)} = 2^7 = \mathbf{128 \text{ grains per square inch at } 100\times}$. (A published G7 would give $N = 2^6 = 64$.)

Q3. Predicted UTS $\approx 3.3 \times 170 = \mathbf{561 \text{ MPa}}$, versus the stated actual UTS of $\sim 750 \text{ MPa}$ — a noticeably low prediction. This is a reasonable finding to report: the empirical 3.3 multiplier is an approximation with real scatter (commonly cited in the roughly 3.0–3.6 range depending on steel grade and condition), and a single published HV value may not represent the exact heat-treatment condition tested for UTS — the correlation is a useful estimate, not a precise conversion.

Q4. Refining Specimen B's grain size would be expected to **increase** toughness — smaller grains mean more grain-boundary area, which both raises yield strength (Hall-Petch) and impedes crack propagation by forcing a growing crack to repeatedly change direction/re-initiate at each new grain, generally improving Charpy impact toughness alongside strength.

Step 3 — Critical Thinking

C1. Since $HV = 1.854F/d^2$, measuring d 10% too large means d^2 is $1.21\times$ the true value ($1.1^2 = 1.21$), so the calculated HV comes out **too low**, by roughly a factor of $1/1.21 \approx 0.83$ — about **17% low**.

C2. The $UTS \approx 3.3 \times HV$ correlation was calibrated empirically to steel's typical microstructure and strain-hardening behavior. Aluminum and brass have entirely different crystal structures, strengthening mechanisms (no pearlite/carbide network), and strain-hardening behavior, so the same multiplier — tuned specifically to steel — would not reliably predict UTS from HV for those alloy systems; each material family has its own empirical HV-to-UTS relationship.

MODULE 7 · WEEK 8 | NDT Methods (Specimens A — 1018, L — 360 leaded brass, P — Ti-6Al-4V)

Representative Data (Step 1): Ti-6Al-4V: ~6% Al, ~4% V, alpha-beta titanium alloy, widely used in aerospace structural and rotating components where in-service NDT (rather than destructive sectioning) is standard practice.

Step 2

Q1. **UT (ultrasonic testing)** is generally the best-positioned NDT method to characterize a particle's size, shape, and depth in detail, but the underlying limitation applies regardless of method: NDT detects a discontinuity by its acoustic, magnetic, or surface signature, not its composition or intent. An engineered lead particle and an unwanted inclusion of similar size and reflectivity can produce very similar NDT indications — distinguishing "intentional" from "defect" reliably usually requires destructive metallography or comparison against a known-good statistical baseline for that alloy.

Q2. MT indication strength depends on orientation relative to the magnetic flux field. Elongated MnS inclusions aligned with the rolling direction — typically parallel to the flux direction used in standard longitudinal magnetization — tend to leak comparatively little flux and would likely produce a **weak** indication; multi-directional magnetization would be needed to reliably detect features aligned this way.

Q3. Titanium aerospace components like Specimen P are extremely expensive, safety-critical, and often irreplaceable once installed in a fielded assembly. Destructive sectioning removes a part permanently from service. NDT (UT, eddy current, radiography) allows repeated in-service inspection throughout the component's life without sacrificing it, so it's used for ongoing monitoring while destructive examination is reserved for special cases.

Q4. Two justified uses: (1) **failure investigation** — determining the root cause of an in-service failure requires destructive sectioning to examine the actual fracture surface and adjacent microstructure; (2) **process qualification / first-article inspection** — verifying that a new production process or heat lot meets microstructural specification before it's trusted for ongoing production, which requires a destructive baseline check.

Step 3 — Critical Thinking

C1. A higher-frequency transducer should be used to detect a smaller defect (shorter wavelength gives finer resolution, since $d_{min} \approx \lambda/2$). The trade-off is **reduced penetration depth** — higher frequencies attenuate more quickly in the material, so a thick section may not be inspectable at the frequency needed to resolve very small defects.

C2. Two specimens could show a very similar UT signature (similar echo amplitude/time-of-flight, reflecting similar bulk acoustic impedance) while having quite different grain size, phase distribution, or fine inclusion content — UT primarily responds to impedance mismatches large enough to reflect sound (voids, cracks, larger inclusions), not to fine microstructural features like grain size or phase morphology, which only metallography resolves directly.

MODULE 8 · WEEK 9 | Quality Standards & Defect Recognition (Specimen A — 1018)

Representative Data (Step 1): Typical published ASTM E112 grain size for normalized 1018 commonly falls in the G7–G8 range; clean, normalized low-carbon steel typically shows thin-series Type A (sulfide) inclusions with little to no Type B/C/D content.

Step 2

Q1. A typical/published grain size number for normalized 1018 is **G ≈ 7–8**; record as **G = 8** (or the specific value found and cited).

Q2. Typical clean, normalized 1018 inclusion rating (ASTM E45):

Type	Description	Typical Rating
A — Sulfide	Gray, elongated (MnS)	Thin, series 1–2
B — Alumina	Angular, clustered	None to trace
C — Silicate	Gray, deformable	None to trace
D — Globular Oxide	Black, round, undeformed	None to trace

Q3. Yes — at G = 8 ($\approx 22 \mu\text{m}$ grain diameter), normalized 1018 typically meets and exceeds a "G7 minimum" certificate requirement (a higher ASTM grain-size number corresponds to finer grain, so G8 is finer than the G7 minimum threshold and passes).

Q4. Four required items: (1) specimen/heat or lot traceability identification, (2) magnification used, (3) the grain-size number and method (comparison chart or intercept method per E112), (4) examiner name/date and the written procedure or standard referenced.

Step 3 — Critical Thinking

C1. A higher ASTM grain size number (n) corresponds to a **smaller/finer** grain, since $N = 2^{(n-1)}$ grows exponentially with n (more grains per unit area means each grain is smaller). Reconciling with Hall-Petch: a higher n (finer grain) means a **stronger** material.

C2. Reproducibility matters more in a quality-standards context because certification/acceptance decisions carry real consequences — part rejection or acceptance, liability, safety — and require a defensible, objective measurement another inspector could reproduce. In a research setting exploring a general trend, an approximate estimate is acceptable because no pass/fail decision rides on the precise reproducibility of the number.

MODULE 9 · WEEK 10 | Wrought Aluminum Alloy Systems (Specimens G — 6061-T6, J — 2024-T3)

Representative Data (Step 1): 2024: ~3.8–4.9% Cu, ~1.2–1.8% Mg, forms $\text{Al}_2\text{Cu}/\text{Al}_2\text{CuMg}$ (S-phase) constituent particles that tend to be coarser than 6061's Mg_2Si ; 2024 is commonly sold clad with pure aluminum for corrosion protection.

Step 2

Q1. 2024 (Al-Cu) is known for **coarser, more numerous** constituent particles than 6061 (Al-Mg-Si) — this matches expectations, since Al-Cu intermetallic phases (Al_2Cu , Al-Cu-Mg-Fe-Mn) tend to form larger, more visible constituent particles during casting/homogenization than the finer Mg_2Si strengthening precipitates typical of 6xxx alloys.

Q2. 2024's higher copper content (~4%) forms Cu-rich intermetallic particles that are electrochemically cathodic relative to the surrounding aluminum matrix, creating galvanic corrosion cells that promote localized pitting and intergranular attack. 6061's Mg-Si chemistry doesn't create the same galvanic couples, giving it inherently better corrosion resistance — which is why 2024 sheet is often sold clad ("Alclad") with a thin layer of pure aluminum for protection.

Q3. No — each alloy's aging response depends on its specific solute chemistry and precipitation sequence (6061: GP zones $\rightarrow \beta'' \rightarrow \beta'$ Mg_2Si ; 2024: GP zones $\rightarrow S'' \rightarrow S'$ Al_2CuMg). Applying 2024's aging schedule to 6061's composition (or vice versa) wouldn't produce the intended precipitate sequence or kinetics for the "foreign" solute system present, so neither alloy would reach its proper peak strength.

Q4. **Fracture toughness / damage tolerance** is a strong answer — 2024-T3 is well known for superior fracture toughness and fatigue-crack-growth resistance, which is exactly why it's specified for aircraft fuselage skins, while 6061 offers better weldability and general structural versatility. An engineer would choose 2024 where damage tolerance/fatigue performance is critical, and 6061 for general-purpose, weldable, corrosion-resistant structural use.

Q5. G (95 HV): $d_{\text{avg}} = \sqrt{(1.854/95)} \approx \mathbf{0.140 \text{ mm}}$. J (125 HV): $d_{\text{avg}} = \sqrt{(1.854/125)} \approx \mathbf{0.122 \text{ mm}}$. J reads harder. This does *not* directly track the coarse constituent-particle density compared in Q1 — those constituent particles are largely insoluble impurity-type intermetallics, not the fine strengthening precipitates. J's higher hardness instead reflects a more effective fine-scale age-hardening response (Al_2CuMg /GP-zone precipitation plus retained cold work in the T3 condition), a separate microstructural feature from the coarse particles discussed in Q1.

Step 3 — Critical Thinking

C1. Both d (grain diameter) and L (precipitate spacing) represent the **average spacing between obstacles** that block dislocation motion — smaller spacing in either case means more obstacles per unit distance, so both terms appear as a spacing measure in the denominator ($\sqrt{d}$ or L) of their respective strengthening equations, just at different length scales (grain-level vs. precipitate-level).

C2. A well-aged (peak, T6) alloy is expected to have somewhat larger, semi-coherent precipitates at the spacing that maximizes Orowan strengthening; a slightly under-aged alloy would have **smaller, more closely-spaced, still largely coherent** precipitates that haven't fully grown yet. The two features expected to differ are precipitate **size** and interparticle **spacing (L)**.

C3. 2024's copper solute has relatively high diffusivity even at room temperature, allowing meaningful natural-aging precipitation (GP zones) without an elevated-temperature step — hence T3/T4 tempers. 6061's Mg_2Si precipitation kinetics are much slower at room temperature, so natural aging alone doesn't reach a useful strength in a practical timeframe — 6061 requires an artificial (elevated-temperature) aging step, hence T6.

MODULE 10 · WEEK 11 | Aluminum Alloys: Heat-Treatable vs. Weldable Structural Grades (Specimens G — 6061-T6, I — 5086)

Representative Data (Step 1): 5086: ~3.5–4.5% Mg, non-heat-treatable, strengthened by Mg in solid solution plus strain hardening (H-temper); widely specified for welded marine aluminum structures over 6xxx alloys.

Step 2

Q1. G (6061-T6) shows a distinct **Mg₂Si precipitate phase** from aging; I (5086) shows **no precipitate phase** — its strengthening comes entirely from Mg dissolved in solid solution plus retained cold work, so its microstructure lacks the second-phase strengthening particles visible in G.

Q2. A welded 6061-T6 joint's heat-affected zone would be expected to show **coarsened, overaged Mg₂Si precipitates** compared to the fully T6-aged base metal — the weld thermal cycle re-dissolves and/or coarsens the fine precipitate structure, producing a locally softened band with fewer, larger, more widely-spaced precipitates than the base metal.

Q3. Cold-formed 5086 would be expected to show **elongated, deformed grains** along the working direction and **increased dislocation density** compared to the as-received condition — the classic strain-hardening/cold-work microstructure.

Q4. **5086** is the better choice for the welded marine bracket — as a non-heat-treatable, solid-solution-strengthened alloy, its strength isn't disrupted by a weld thermal cycle the way 6061-T6's precipitate structure is. 6061-T6 would soften significantly in the HAZ after welding, which is exactly why 5xxx alloys dominate welded marine aluminum applications.

Q5. G (95 HV): $d_{avg} \approx 0.140 \text{ mm}$ (from Module 9). I (70 HV): $d_{avg} = \sqrt{(1.854/70)} \approx 0.163 \text{ mm}$. This shows 5086 is noticeably softer than 6061-T6 (about 26% lower HV) but not dramatically so — precipitation hardening gives G a meaningfully larger strength boost than solid-solution strengthening alone gives I, though 5086 remains a solidly capable structural alloy on its own.

Step 3 — Critical Thinking

C1. A higher strain-hardening exponent n favors cold-forming because the metal keeps resisting further localized thinning as it deforms, delaying the onset of necking and allowing more uniform strain distribution during forming. A low- n metal reaches its work-hardening limit quickly and is prone to premature localized necking during deep forming operations.

C2. This shows that "high-strength" and "weld-ready" are two independent selection criteria — an alloy optimized for peak as-delivered strength (6061-T6's precipitation hardening) can underperform after a weld thermal cycle, while an alloy that never had a high peak strength to lose (5086) retains its strength through welding. A designer must match the selection criterion to the actual service condition (unwelded high-strength part vs. welded structure), not assume the strongest as-delivered alloy is automatically the best choice once welding enters the picture.

MODULE 11 · WEEK 12 | Machining Processes (Specimens A — 1018, B — 1045, G — 6061 aluminum)

Representative Data (Step 1): Published machinability ratings commonly cite 1018 as having good machinability (often used as a reference/baseline grade), 1045 as moderate (higher carbon increases cutting forces and tool wear versus 1018), and 6061 aluminum as excellent machinability but with higher tendency toward built-up edge without proper tooling/coolant.

Step 2

Q1. Machining of 1018 typically produces a thin, plastically deformed/work-hardened surface layer with grains sheared and elongated in the direction of tool travel near the machined surface, transitioning quickly to the undeformed, equiaxed bulk grain structure just below — visible as a distinct deformed layer under magnification.

Q2. Increased cutting forces and heat from tool wear can locally reheat the surface enough to alter or temper the near-surface microstructure, and in more severe cases cause a thin "**white layer**" — untempered, hard, brittle martensite formed by frictional heating followed by rapid self-quenching from the surrounding bulk material. It appears as a thin, featureless, white-etching band under magnification.

Q3. A machining-induced **white layer** (or a heavily deformed/microcracked near-surface layer) is hard, brittle, and can carry residual tensile stress — any of these can act as a stress concentrator and fatigue-crack initiation site in service.

Q4. **Aluminum (G)** typically shows more machining-induced near-surface deformation than the steels. Aluminum's lower strength and stiffness mean the cutting-zone shear deformation extends more readily and deeper into the material before chip formation occurs, making it more prone to a thicker deformed/smeared surface layer relative to its bulk hardness.

Q5. HV: A = 125, B = 170, G = 95. Ranked softest → hardest: **G (95) < A (125) < B (170)**. Harder materials generally require lower cutting speeds and cause more tool wear per the Taylor relationship, so **B (1045, hardest)** would need the lowest cutting speed and show the most tool wear, while **G (6061, softest)** could be cut fastest with the least tool wear.

Step 3 — Critical Thinking

C1. Increasing cutting speed V **decreases** tool life T ($V \cdot T^n = C$ means V and T are inversely related for fixed C, n). A harder specimen (higher HV) generates more cutting force and heat at a given speed, so it would need a **lower** cutting speed to achieve the same tool life as a softer specimen.

C2. Hard, abrasive second-phase particles or inclusions (coarse carbides, oxide/silicate inclusions) can accelerate tool wear through abrasive action independent of the workpiece's bulk hardness. Two specimens with identical bulk HV could still machine very differently if one contains abrasive inclusions the Taylor equation's V and T terms don't directly capture.

MODULE 12 · WEEK 13 | Copper Alloys: Strengthening Mechanisms (Specimens N — 110 copper, O — 182 chromium copper)

Representative Data (Step 1): 110 copper (C11000): 99.90% min Cu, excellent electrical conductivity (~101% IACS). 182 chromium copper (C18200): ~0.6–1.2% Cr, precipitation-hardenable, conductivity typically ~80–85% IACS after aging — a real but modest reduction from pure copper.

Step 2

Q1. N (pure copper) shows a simple, single-phase equiaxed grain structure with prominent annealing twins (very common in pure FCC copper's low stacking-fault energy) and no second-phase particles. O (chromium copper) shows a broadly similar twinned matrix but with fine, dispersed Cr-rich precipitates from aging. O is "less simple" because it's an alloy — pure single-element metals like N inherently lack the solute-clustering/precipitation complexity that alloying introduces.

Q2. Precipitate phases: F (17-4PH) → Cu-rich ϵ -phase clusters in a martensitic matrix; J (2024-T3) → Al_2CuMg (S-phase) plus GP zones in an FCC aluminum matrix; O (182 Cr-Cu) → nearly pure chromium precipitates in an FCC copper matrix. Common to all three: each forms a fine, coherent or semi-coherent second phase from a supersaturated solid solution via a solutionize → quench → age sequence, and all strengthen through the same Orowan/precipitate-pinning mechanism of blocking dislocation motion.

Q3. Any solute atom, even in small amounts, locally distorts the host lattice and scatters conduction electrons (adds an impurity-scattering contribution per Matthiessen's rule) — this raises resistivity regardless of whether that solute also strengthens the material. A pure, unalloyed metal has the most orderly lattice and least electron scattering, hence the highest conductivity of any alloy in its family.

Q4. A PM (sintered) part would be expected to show **residual porosity** — small voids between the original powder particles that never fully densified during sintering — a feature essentially absent in the wrought, fully-dense alloys researched throughout the rest of the semester.

Q5. N (50 HV): $d_{\text{avg}} = \sqrt{(1.854/50)} \approx \mathbf{0.193 \text{ mm}}$. O (130 HV): $d_{\text{avg}} = \sqrt{(1.854/130)} \approx \mathbf{0.119 \text{ mm}}$. The jump from N to O is 50 → 130 HV, a **2.6× increase** — proportionally similar in scale to the aluminum precipitation-hardening jump (annealed 6061 ≈ 30 HV → T6 ≈ 95 HV, roughly 3.2×), and clearly smaller than steel's jump (annealed 304 ≈ 150 HV → aged 17-4PH ≈ 375 HV, ≈2.5× but from a much higher starting point and driven by both a martensitic transformation *and* precipitation together). The N→O jump looks more like the aluminum case — precipitation hardening alone, without an accompanying phase transformation.

Step 3 — Critical Thinking

C1. O has higher resistivity than N because O's chromium solute/precipitate atoms contribute a nonzero impurity term (electron scattering from alloying), while N (pure copper) has impurity ≈ 0. At the same test temperature, ρ_{thermal} is essentially equal for both (same base copper lattice), so the full difference in ρ_{total} comes from the added impurity term in O.

C2. The same precipitates that pin dislocations (via lattice strain/coherency mismatch around each particle) also scatter conduction electrons, the same way solute atoms or lattice defects do — the microstructural feature responsible for blocking dislocation motion (raising strength) is the same feature responsible for scattering electrons (raising resistivity). The two effects can't be decoupled; the trade-off is intrinsic to using a second phase to strengthen a highly conductive metal like copper.

MODULE 13 · WEEK 14 | Brass Alloys: Composition and Microstructure (Specimens K — 260 brass, L — 360 brass, M — 464 brass)

Representative Data (Step 1): 260 (cartridge): ~70% Cu, ~30% Zn, single-phase alpha. 360 (lead, free-machining): ~60–63% Cu, ~35–37% Zn, ~2.5–3.5% Pb. 464 (naval): ~59–62% Cu, ~37–40% Zn, ~0.5–1% Sn, two-phase alpha-beta. The alpha-phase solubility limit for zinc in the Cu-Zn system is commonly cited around ~35% Zn at room temperature.

Step 2

Q1. Annealing twins remain clearly visible in single-phase alpha brass **K**; in lead brass **L**, twins in the surrounding alpha matrix are still generally visible since lead simply sits as discrete globules without disrupting the matrix itself; in duplex alpha-beta brass **M**, twins are harder to distinguish because the high-contrast alpha/beta phase boundaries dominate the visual field and the beta phase (an ordered BCC-derived structure) doesn't show the same FCC-style annealing twins.

Q2. Lead is essentially insoluble in the Cu-Zn matrix, so it exists as discrete, soft, low-melting-point globules distributed through the microstructure rather than in solid solution. During machining, these globules act as internal chip-breakers and lubricant pockets that promote clean chip fracture and reduce cutting forces — a benefit that depends entirely on lead staying separate from (not dissolving into) the matrix.

Q3. Since K (~30% Zn) is single-phase alpha and M (~38–39% Zn) is two-phase alpha-beta, the alpha/beta phase boundary must fall **between roughly 30% and 38–39% Zn** — consistent with the commonly cited ~35% Zn solubility limit of the alpha phase at room temperature.

Q4. This is a genuine trade-off: **360** machines more easily and cheaply (lead, single-phase alpha, excellent machinability) but its zinc content and any beta-phase presence increase susceptibility to dezincification in seawater; a duplex or dezincification-resistant naval-type brass (with a small As/Sb/P inhibitor addition) sacrifices some machinability for long-term seawater durability. A designer prioritizing service life in seawater would typically specify the naval/duplex option despite the added machining cost.

Q5. K (85 HV): $d_{avg} = \sqrt{(1.854/85)} \approx \mathbf{0.148\text{ mm}}$. L (80 HV): $d_{avg} = \sqrt{(1.854/80)} \approx \mathbf{0.152\text{ mm}}$. M (90 HV): $d_{avg} = \sqrt{(1.854/90)} \approx \mathbf{0.144\text{ mm}}$. M does read modestly harder than K and L, consistent with the added beta phase (a harder, more brittle ordered structure) contributing extra hardness beyond single-phase alpha — though the margin is fairly small at these reference values, so it's a real but modest confirmation of the phase-count trend.

Step 3 — Critical Thinking

C1. For a beta phase to appear at all, M's overall zinc content (C_0) must exceed the alpha phase's maximum zinc solubility limit (C_α , ~35% at room temperature) — i.e., C_0 must fall to the right of C_α on the Cu-Zn diagram for the lever rule to give a nonzero beta weight fraction.

C2. No — lead does not appear anywhere in the binary Cu-Zn lever-rule calculation; it's a third element entirely outside the two-component system the phase diagram describes. This illustrates a real limitation of a simple binary phase diagram: it only accounts for the two plotted elements, so any additional alloying addition (lead in L, tin in M) and its effects (machinability, dezincification resistance) must be understood from separate data — a true multicomponent phase diagram or empirical data would be needed to capture those effects quantitatively.

MODULE 14 · WEEK 15 | Precipitation Hardening: A Cross-Family Comparison (Specimens F — 17-4PH, J — 2024-T3, O — 182 chromium copper)

Representative Data (Step 1): F: martensitic matrix, Cu-rich ϵ -phase precipitate. J: FCC aluminum matrix, Al_2CuMg (S-phase)/GP-zone precipitates. O: FCC copper matrix, chromium-rich precipitates. All three follow the same solutionize $\rightarrow$ quench $\rightarrow$ age sequence.

Step 2

Q1. Table:

Specimen	Base Metal	Precipitate Phase	Typical Scale
F	17-4PH stainless (martensitic)	Cu-rich ϵ -phase	Fine, hard to resolve optically
J	2024-T3 aluminum (FCC)	Al_2CuMg (S-phase) / GP zones	Fine, hard to resolve optically
O	182 chromium copper (FCC)	Chromium-rich precipitates	Fine, hard to resolve optically (properly aged)

Q2. **Solutionize:** heat into a single-phase field to dissolve solute into a homogeneous solid solution, erasing any prior precipitate structure. **Quench:** rapidly cool to trap the solute in a supersaturated solid solution, preventing slow, coarse precipitation on cooling. **Age:** hold at an intermediate elevated temperature to allow controlled, fine-scale precipitation of the strengthening phase. Using O as the example: solutionizing dissolves Cr into the Cu matrix, quenching traps it in supersaturated solution, and aging at an elevated temperature precipitates the fine Cr particles that pin dislocations.

Q3. Fine, closely-spaced precipitates force a dislocation to bow with tighter curvature to pass between them (Orowan looping), and the stress required for that bowing is inversely proportional to interparticle spacing ($\Delta\sigma \approx Gb/L$). Coarse, widely-spaced precipitates — even at the same total volume fraction — require much less stress to bypass, so they're less effective obstacles.

Q4. No — natural (room-temperature) aging would generally not reach the same strength as artificial (elevated-temperature) aging within a practical timeframe. Diffusion is strongly temperature-dependent (Arrhenius behavior), so at room temperature solute diffusion is far slower — precipitation either stalls at an early stage or takes impractically long. 2024's Cu-Mg system does naturally age usefully at room temperature due to relatively high solute mobility, but even it doesn't reach the peak strength achievable with an elevated-temperature aging step.

Q5. F (375 HV): $d_{\text{avg}} = \sqrt{(1.854/375)} \approx \mathbf{0.0703 \text{ mm}}$. Comparing relative (proportional) strength increase from precipitation hardening using the given baselines: steel $\approx 375/150 \approx \mathbf{2.5\times}$; aluminum $\approx 125/30 \approx \mathbf{4.2\times}$; copper $\approx 130/40 \approx \mathbf{3.25\times}$. By this relative measure, **aluminum (J) gained the most**, though steel's *absolute* HV increase (225 points) is the largest in absolute terms — a good discussion point about the difference between relative and absolute comparisons.

Step 3 — Critical Thinking

C1. No — a higher G alone doesn't guarantee the largest $\Delta\sigma$, because Orowan strengthening ($\Delta\sigma \approx Gb/L$) depends equally on interparticle spacing L . Even with roughly 3× higher shear modulus, if F's precipitate spacing is much larger (coarser, more widely-spaced) than J's or O's fine precipitates, that larger L could offset or overcome G 's advantage. You'd need each alloy's actual measured precipitate spacing to answer with confidence.

C2. F's actual HV (375) is dramatically higher than either J (125) or O (130) — far beyond what a roughly 3× shear-modulus advantage alone would predict if precipitate spacing were similar across all three. This suggests either F's precipitate spacing is also more favorable, or — more likely — F's overall hardness reflects more than precipitation hardening alone: its underlying martensitic matrix is inherently much harder than the FCC aluminum or copper matrices in J and O. Base-matrix strength differences appear to dominate over the Orowan G -term alone in explaining the real HV rankings.

MODULE 15 · WEEK 16 | Course Capstone & Synthesis

Step 2

Q1. Example structures: **BCC** — ferrite in Specimen A (1018 steel); **FCC** — Specimen D (304 austenitic stainless steel) or Specimen N (110 copper); **HCP** — Specimen P (Ti-6Al-4V, alpha-phase titanium is HCP).

Q2. FCC metals have close-packed {111} slip planes with relatively low stacking-fault energy, giving a low-energy pathway for stacking faults (twins) to form during annealing. BCC carbon steels' more open crystal structure lacks that same low-energy twinning pathway under normal annealing conditions, so annealing twins are essentially absent in BCC ferrite even though mechanical/deformation twinning can occur in BCC metals under very different conditions (high strain rate, low temperature).

Q3. Example: **Specimen I (5086 aluminum)** is strengthened primarily by **solid-solution strengthening** — magnesium dissolved substitutionally in the aluminum lattice distorts it and impedes dislocation motion. As a non-heat-treatable alloy, it has no precipitation-hardening contribution; Hall-Petch grain-boundary strengthening also contributes to some degree, as it does in any polycrystalline metal.

Q4. Unit 1 (atomic bonding: metallic, non-directional bonding is what allows grain boundaries to exist) → Unit 2 (alloy family/composition: carbon content or alloying additions like Mo or Cu determine which phases and precipitates form — e.g., Specimen B's higher carbon produces more pearlite; Specimen E's Mo addition raises PREN) → Unit 3 (mechanical testing/inspection: those same phases are exactly what a Vickers indent or grain-size measurement quantifies — e.g., Specimen F's aged martensitic structure gives it by far the highest logged HV) → Unit 4 (manufacturing/alloy selection: that same microstructure governs process behavior — e.g., Specimen G's Mg₂Si precipitates soften in a weld HAZ, which is why Specimen I is preferred for welded structures) → Unit 5 (service/corrosion: composition again governs long-term performance — Specimen E's Mo addition, the same chemistry lever from Unit 2, is exactly why it resists chloride attack better in service). At least three specimens (B, F, G/I, E) cited as evidence.

Q5. Compiled Hardness Log, softest to hardest: **N (50) < I (70) < L (80) < K (85) < M (90) < G (95) < A (125) ≈ J (125) < O (130) < B (170) < F (375)**. Example pair: **I (5086 aluminum, 70 HV)** and **L (360 brass, 80 HV)** — despite being different base metals (aluminum vs. a copper alloy) with different crystal structures, their hardness values land close together because both derive their strength primarily from **solid-solution strengthening alone** (Mg in Al for I; Zn in Cu for L), without precipitation hardening or a duplex second phase reinforcing either one — a comparable, modest strengthening mechanism in both cases produces comparable mid-range hardness.

Step 3 — Critical Thinking

C1. A metal's corrosion resistance (PREN, governed by Cr/Mo/N content and passive-film chemistry) and its strengthening mechanism (grain size, precipitates, dislocation density) are governed by largely independent aspects of composition. Using D vs. E (304 vs. 316): both are annealed austenitic stainless steels with essentially the same strengthening mechanism and similar hardness (≈150–160 HV), yet 316's Mo addition dramatically raises its PREN/corrosion resistance without meaningfully changing its strengthening mechanism or crystal structure — showing a metal can be "re-engineered" for corrosion resistance largely independent of its strength.

C2. Most directly connected to a visual/microstructural observation: **Hall-Petch** — grain size is directly visible and measurable under a microscope, and its effect on strength is intuitive from what you see (Specimens A/B, Module 1). Hardest to connect to a visual observation: **PREN** — corrosion resistance depends on the chemistry of a passive oxide film operating at a scale far below optical (even most electron-microscope) resolution; you cannot "see" why one alloy resists pitting better the way you can see a twin or a precipitate (Specimens D/E, Module 4).